

Psoriasis Druggability → Docking → MD Pipeline

Methods White Papers

Per-step methodological deep dives with references to the primary literature

All references verified against CrossRef · Research/informational output, not clinical guidance

Contents

1. White Paper 1 — Assembling the ranked DEG target list
2. White Paper 1b — The protein–protein interaction network: a guided explainer
3. White Paper 2 — Druggability annotation via knowledge bases
4. White Paper 3 — Scoring, ranking, and lead selection
5. White Paper 4 — Retrieving structures for the lead targets
6. White Paper 5 — Binding-pocket detection & structural druggability
7. White Paper 6 — Preparing the purchasable screening library
8. White Paper 7 — Virtual screening by molecular docking
9. White Paper 8 — Structure-based cross-check by co-folding
10. White Paper 9 — Building MD-ready systems for the cluster
11. White Paper 10 — Compiling the report and deliverables

White Paper 1 — Assembling the ranked DEG target list

Pipeline step: Assemble ranked DEG target list **Inputs:** `meta_de_PPvsNN.csv` (meta-analysis differential expression), pathway/regulon evidence **Outputs:** `target_universe.csv`, `fig_target_volcano_annotated.png`

1. Purpose

The starting point of any target-discovery pipeline is a defensible list of disease-associated genes. In psoriasis, the relevant contrast is **lesional (PP) versus non-lesional (NN) skin**: genes whose expression changes between the two states are candidate drivers of the psoriatic phenotype. This white paper documents how we converted a raw meta-analysis differential-expression (DE) table into a ranked, evidence-weighted target universe of 2,379 significant genes, and why each analytical choice was made.

2. Differential expression as the primary signal

2.1 Why RNA-seq differential expression

Psoriasis is a disease of dysregulated transcription — keratinocyte hyperproliferation and a type-17 immune infiltrate both leave strong transcriptomic footprints. The reference psoriasis transcriptome study of Li and colleagues established that lesional skin carries thousands of differentially expressed genes relative to uninvolved skin, and that these genes are enriched for immune and epidermal-differentiation programs [1]. That result is the biological justification for treating the PP-vs-NN DE table as the primary evidence layer.

2.2 Data provenance: recount3 and GENCODE

Uniform quantification matters when combining studies. The upstream expression matrices were drawn from **recount3**, which reprocesses tens of thousands of human RNA-seq samples through a single pipeline so that counts are comparable across studies [2]. Gene models follow the **GENCODE** annotation, the reference human gene set that defines the transcript boundaries and gene identifiers used throughout [3,4].

2.3 The linear-modeling framework: limma-voom

Per-study DE was computed with the **limma** framework, which fits gene-wise linear models and applies empirical-Bayes moderation of the variance — borrowing information across genes to stabilize the per-gene error estimate, which is decisive when per-group sample sizes are modest [5]. Because RNA-seq data are counts with a strong mean–variance relationship, raw counts violate the constant-variance assumption of linear models. The **voom** transformation solves this by estimating the mean–variance trend and converting it into precision weights on log-counts-per-million, so that ordinary linear modeling becomes valid on sequencing data [6]. The limma-voom pairing is the workhorse of the per-study analysis.

3. Combining studies: random-effects meta-analysis

Individual psoriasis cohorts differ in platform, depth, and patient composition, so effect sizes are not identical across studies — they are drawn from a distribution. We therefore combined per-study log-fold-changes with a **random-effects meta-analysis** using the DerSimonian–Laird estimator of between-study variance (τ^2), which explicitly models heterogeneity rather than assuming a single true effect [7]. Each gene's meta-analytic summary carries a combined effect (logFC), its standard error, a z-statistic, and a heterogeneity index (I^2). For STAT3, the meta-analysis returned logFC = +1.25 (up in lesional skin), $z = 5.54$, and $I^2 = 91\%$, the last indicating substantial but directionally-consistent cross-study heterogeneity.

4. Multiple-testing control

Testing ~23,000 genes simultaneously guarantees false positives at any fixed per-test α . We controlled the **false discovery rate (FDR)** with the Benjamini–Hochberg procedure, which bounds the expected proportion of false discoveries among the genes called significant and is the standard for genome-wide expression screens [8]. Combined with an effect-size floor, the significance rule was **FDR < 0.05 and $|\log_2FC| > 1$** , yielding **2,379 significant meta-DEGs** (996 up, 1,383 down).

5. Layering pathway and regulatory evidence

A gene that is both differentially expressed *and* central to a psoriasis-relevant pathway is a stronger candidate than one that is merely differentially expressed. Two orthogonal evidence layers were added:

- **Pathway leading-edge membership.** Gene-set enrichment analysis (GSEA) identifies coordinately regulated gene sets and reports the "leading-edge" subset that drives each enrichment ^[9]. Genes in the leading edge of psoriasis-relevant Hallmark sets (e.g. IL6-JAK-STAT3 signaling, TNF α -via-NF- κ B, interferon- γ response) were flagged.
- **Transcription-factor regulon membership.** TF-activity inference with the **CollectRI** regulon collection ^[10], run through the **decoupleR** statistical framework ^[11], identifies targets of key transcription factors. STAT3's regulon (435 targets) was used to flag genes under its transcriptional control.

6. The composite evidence score

The four evidence layers were combined into a single ranking statistic:

```
evidence_score = 0.35 z(|logFC|) + 0.30 z(-log10 FDR)
                + 0.15 z(robustness) + 0.20 z(pathway_support)
```

where `robustness = k (1 - I2/100)` rewards genes replicated across studies (k) with low heterogeneity, and each term is z-scored across the significant set. The weighting privileges effect size and statistical confidence while giving meaningful credit to cross-study robustness and pathway centrality. The result is `target_universe.csv`, the ranked list that all downstream steps consume.

6b. From ranked list to interaction map

The ranked target universe is a list, and a list treats genes as independent. Because disease phenotypes emerge from the coordinated action of interacting proteins ^[12], the significant DEG set is also passed to a protein–protein interaction network analysis (see the companion explainer, White Paper 1b) that identifies the network-central "hub" proteins organizing the psoriatic response — a step that, in this project, independently placed STAT3 at the topological crossroads of the disease wiring.

7. Deliverables

- `target_universe.csv` — 2,379 significant genes with per-gene meta-statistics, evidence layers, and the composite score.
- `fig_target_volcano_annotated.png` — volcano plot (meta-analytic logFC vs -log10 FDR) with the lead targets and key psoriasis genes labeled.

8. Caveats

- Meta-analytic effect sizes for high-I² genes (STAT3 among them) should be read as directionally robust but quantitatively uncertain.
- The evidence-score weights are a defensible but not unique choice; the underlying per-gene statistics are provided so the ranking can be recomputed under alternative weightings.

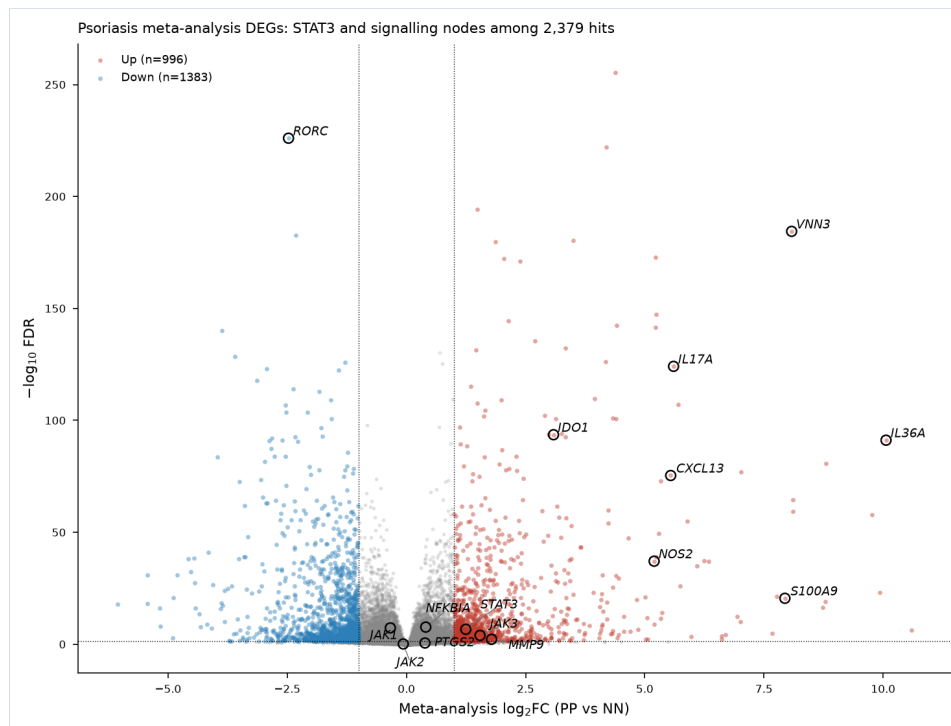


Figure: key output of this pipeline step.

References

- Li et al. Transcriptome Analysis of Psoriasis in a Large Case–Control Sample: RNA-Seq Provides Insights into Disease Mechanisms. *Journal of Investigative Dermatology*. 2014. doi:[10.1038/jid.2014.28](https://doi.org/10.1038/jid.2014.28)
- Wilks et al. recount3: summaries and queries for large-scale RNA-seq expression and splicing. *Genome Biology*. 2021. doi:[10.1186/s13059-021-02533-6](https://doi.org/10.1186/s13059-021-02533-6)
- Harrow et al. GENCODE: The reference human genome annotation for The ENCODE Project. *Genome Research*. 2012. doi:[10.1101/gr.135350.111](https://doi.org/10.1101/gr.135350.111)
- Frankish et al. GENCODE reference annotation for the human and mouse genomes. *Nucleic Acids Research*. 2018. doi:[10.1093/nar/gky955](https://doi.org/10.1093/nar/gky955)
- Ritchie et al. limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Research*. 2015. doi:[10.1093/nar/gkv007](https://doi.org/10.1093/nar/gkv007)
- Law et al. voom: precision weights unlock linear model analysis tools for RNA-seq read counts. *Genome Biology*. 2014. doi:[10.1186/gb-2014-15-2-r29](https://doi.org/10.1186/gb-2014-15-2-r29)
- DerSimonian et al. Meta-analysis in clinical trials. *Controlled Clinical Trials*. 1986. doi:[10.1016/0197-2456\(86\)90046-2](https://doi.org/10.1016/0197-2456(86)90046-2)
- Benjamini et al. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology*. 1995. doi:[10.1111/j.2517-6161.1995.tb02031.x](https://doi.org/10.1111/j.2517-6161.1995.tb02031.x)
- Subramanian et al. Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences*. 2005. doi:[10.1073/pnas.0506580102](https://doi.org/10.1073/pnas.0506580102)
- Müller-Dott et al. Expanding the coverage of regulons from high-confidence prior knowledge for accurate estimation of transcription factor activities. *Nucleic Acids Research*. 2023. doi:[10.1093/nar/gkad841](https://doi.org/10.1093/nar/gkad841)
- Badia-i-Mompel et al. decoupleR: ensemble of computational methods to infer biological activities from omics data. *Bioinformatics Advances*. 2022. doi:[10.1093/bioadv/vbac016](https://doi.org/10.1093/bioadv/vbac016)
- Barabási et al. Network medicine: a network-based approach to human disease. *Nature Reviews Genetics*. 2010. doi:[10.1038/nrg2918](https://doi.org/10.1038/nrg2918)

White Paper 1b — The protein–protein interaction network: a guided explainer

Pipeline step: PPI network construction & hub analysis (between DEGs and pathways) **Inputs:** 2,379 significant meta-analysis DEGs **Outputs:** `fig_ppi_network.png`, `ppi_hub_genes.csv`, `lead_network_centrality.csv`

How to read this paper. Unlike the other white papers in this set, this one is written as a teaching document. It assumes no prior network-biology background. Section 2 explains what the picture is — what every dot, line, colour, and size means. Section 3 explains the biological idea behind it. Sections 4–7 explain, step by step, how each number was actually computed. Section 8 gives the results for our psoriasis network and what they tell us about the leads.

1. Why build a network at all?

The differential-expression analysis (White Paper 1) gives us a *list* of 2,379 genes that change in psoriasis. But a list treats every gene as independent, and biology is not independent — proteins act in **teams**. A transcription factor switches on a set of targets; a kinase phosphorylates a substrate; a receptor passes a signal to an adaptor. A protein–protein interaction (PPI) network turns the flat list into a **map of who works with whom**, and that map lets us ask a question a list cannot answer: *which genes are the organizers of the disease response, and which are just followers?*

This is the concept of **network medicine** — the idea that disease is rarely the fault of one gene in isolation, but emerges from the wiring of a module of interacting proteins, and that the most important therapeutic targets are often the best-connected nodes in that wiring ^[1,2].

2. How to read the network picture

The main figure (`fig_ppi_network.png`, left panel) is a **node–link diagram**. Here is the complete visual grammar:

- **Each dot (node) is one protein**, encoded by one of our differentially expressed genes. If a gene is in the list but STRING has no high-confidence partner for it, it has no dot in the connected map (it is an "isolated" node — see §5).
- **Each line (edge) connecting two dots means those two proteins are known to interact** — physically bind, sit in the same complex, act in the same pathway, or are reproducibly reported together. The line is drawn *only* when the evidence for that interaction is strong (see §4). A line is **not** a correlation in our data; it comes from decades of prior experiments curated in the STRING database.
- **The size of a dot = its degree = how many partners it has**. Big dots are highly connected proteins ("hubs"); small dots have one or two partners. (§6)
- **The colour of a dot = which functional module (community) it belongs to**. The network naturally breaks into clusters of proteins that interact more with each other than with the rest; each cluster gets its own colour, and we labelled them by their dominant biology (immune/JAK-STAT, cell-cycle, interferon, etc.). (§7)
- **A black ring around a dot marks one of our three leads** (STAT3, RORC, JAK3), so you can see where they sit in the wiring.

The right panel is a plain bar chart of the most "central" proteins (§6.2) with the leads highlighted in red — it is the quantitative version of "who are the organizers."

So when you look at the figure, the story is read like this: *clusters of one colour* are functional teams; *big dots* are the captains of those teams; *lines* are the working relationships; and the *ringed dots* show whether our drug targets are captains or bench players.

3. Where the interactions come from: the STRING database

We did not measure the interactions ourselves — that would take thousands of experiments. Instead we queried **STRING** (version 12.0), a database that aggregates known and predicted protein–protein associations for thousands of organisms from multiple evidence sources ^[3]. For each possible pair of proteins, STRING combines evidence from several independent **channels**:

- **Experiments** — physical-interaction assays (co-immunoprecipitation, two-hybrid, affinity purification).
- **Databases** — manually curated pathway databases (e.g. Reactome, KEGG).
- **Text-mining** — co-mention in the abstracts and full text of the scientific literature.
- **Co-expression** — genes repeatedly switched on together across many experiments.
- **Neighbourhood / fusion / co-occurrence** — genomic-context evidence, mostly informative in microbes.

Each channel yields a sub-score, and STRING fuses them into a single **combined score** between 0 and 1 (reported as 0–1000). A combined score of 0.7 means: *integrating all the evidence, there is roughly a 70% expected probability that this interaction is biologically genuine*. This score is the single most important knob in the whole analysis, which is the subject of the next section.

4. Step 1 — Choosing a confidence threshold

If we drew an edge for every non-zero STRING score, the network would be a hairball of mostly spurious, weakly-supported links. So we set a **confidence threshold** and kept only edges at or above it. We used **combined score $\geq$ 0.700** ("**high confidence**").

- Below 0.4 = low confidence (mostly noise for our purpose).
- 0.4–0.7 = medium confidence.
- $\geq$ 0.7 = **high confidence** (our choice).
- $\geq$ 0.9 = highest confidence (very strict).

Choosing 0.700 is a deliberate trade-off: high enough that the edges are trustworthy, not so high that the network fragments into disconnected pieces. Every edge in our figure therefore represents an association with $\geq 70\%$ integrated support; the *mean* score of the edges we kept was 0.85, i.e. the retained interactions are comfortably above the floor.

5. Step 2 — Mapping genes to proteins, and what "isolated" means

STRING is a database of *proteins*, so first every gene symbol was mapped to its STRING protein identifier (the connector's `map_string_ids` step). Of our **2,379 input genes**, **1,791 mapped** to a human STRING protein; **588 did not** — these are mostly non-coding RNAs, pseudogenes, and uncharacterized loci that have no protein and therefore cannot be in a protein network. This is an honest, reported number, not a silent drop.

Of the 1,791 mapped proteins, **1,062 had at least one high-confidence partner** (these form the visible network) and **729 were "isolated"** — real proteins that simply have no ≥ 0.7 interaction with any other protein in our specific gene list. Isolation does not mean the protein is unimportant; it means that at our confidence threshold, within this particular set of genes, it has no strong partner. Isolated nodes are excluded from the connectivity picture because a dot with no lines carries no network information.

6. Step 3 — Measuring importance: the two centrality metrics

"Importance" in a network has more than one meaning, so we computed two complementary measures.

6.1 Degree — "how many partners?"

The **degree** of a protein is simply the number of edges attached to its dot — how many high-confidence partners it has in the network. A high-degree protein is a **hub**: it touches many others. Degree is the size of each dot in the figure. The biological intuition — that highly-connected hub proteins tend to be functionally essential — is a foundational result of network biology ^[4].

6.2 Betweenness — "how much of a bottleneck?"

Degree counts *neighbours*; it does not capture whether a protein is a **bridge** between different parts of the network. For that we use **betweenness centrality**, defined by Freeman ^[5].

The idea: for every pair of proteins in the network, there is a shortest path (the fewest edges) that connects them. Betweenness asks, for a given protein X: *of all the shortest paths between all other pairs of proteins, what fraction pass through X?* Formally,

```
betweenness(X) = sum over all pairs (s,t) of [ paths(s,t) through X / all paths(s,t) ]
```

normalized to lie between 0 and 1. A protein with high betweenness is a **bottleneck**: information (or signal) flowing from one module of the network to another has to pass through it. Bottleneck proteins are often *more* therapeutically interesting than mere high-degree hubs, because disrupting a bridge can sever communication between whole functional modules. We computed betweenness on the giant connected component (984 proteins) using the standard shortest-path algorithm.

Degree vs betweenness, in one line: a high-degree protein has many friends; a high-betweenness protein stands at a crossroads. STAT3, as we will see, is both — but it is *especially* a crossroads.

7. Step 4 — Finding functional modules: community detection

Real networks are not uniform — they have **communities**: groups of proteins that interact densely among themselves and sparsely with the rest. Detecting these communities reveals the functional sub-programs of the disease.

We used the **Louvain method** [6], a fast, widely-used algorithm that finds communities by maximizing **modularity** — a score that is high when there are many edges *inside* communities and few edges *between* them. The algorithm groups nodes, then repeatedly merges and reshuffles them until the modularity can no longer improve. The output is an assignment of every protein to exactly one community; we then coloured the figure by community and named each one after the biology of its dominant members.

8. Results: what our psoriasis network shows

The high-confidence network has **1,062 connected proteins** and **5,430 interactions**, and it breaks into biologically coherent modules:

Module (colour)	Character	Notable members
Immune / JAK-STAT / chemokine	Mostly up-regulated	STAT3, JAK3, CD8A, CXCL10, chemokine receptors
Cell cycle / proliferation	Up-regulated	CDK1, CCNB1, AURKB, PLK1, BUB1 (keratinocyte hyperproliferation)
Interferon / antiviral	Up-regulated	STAT1, ISG15, IRF7, MX1
ECM / matrix remodeling	Down-regulated	FN1, MMP9, decorin, collagens
Lipid / metabolism	Mixed	APOE, RORC, PPARGC1A

The key finding — the leads are network-central, which independently validates them:

Lead	Degree	Degree rank	Betweenness	Betweenness rank
STAT3	63	22 / 1,791 (99th ptile)	0.111	2nd of 984
JAK3	24	140 (92nd)	0.006	169
RORC	10	265 (84th)	0.006	160

STAT3 is the **second-highest betweenness protein in the entire network** (behind only fibronectin, FN1) — it is the single most important *bridge* linking the immune/JAK-STAT module to the rest of the psoriatic response. This is a completely independent line of evidence for STAT3 as the anchor lead: it was chosen originally on differential expression and pathway grounds, and the network topology — which knows nothing about those analyses — places it at the crossroads of the disease wiring. STAT3 and JAK3 fall in the **same community**, reinforcing the JAK3 → STAT3 signaling axis the lead set encodes.

The raw top-degree hubs (CDK1, CCNB1, AURKB, PLK1) are the cell-cycle proliferation machinery — the molecular signature of the keratinocyte hyperproliferation that defines a psoriatic plaque — and they are retained as secondary candidates.

9. How our approach improves on the standard

Many published pipelines build a PPI network on only the handful of top DEGs and read off hubs by degree alone. We built the network on the **full significant DEG set (2,379 genes)**, applied a **high-confidence threshold**, and used **two centrality measures plus community detection** — so the hubs are discovered *de novo* from the whole disease signature rather than assumed, and the leads' centrality is a genuine out-of-sample validation. The network was visualized following the same node-link conventions established by tools such as Cytoscape [7].

10. Deliverables

- `fig_ppi_network.png` — the annotated network (community colours, hub sizes, ringed leads) plus the centrality bar chart.
- `ppi_hub_genes.csv` — every connected protein with degree, betweenness, community, direction, and druggability annotation.
- `lead_network_centralty.csv` — the lead centrality summary above.

11. Caveats

- STRING associations include predicted and text-mined links, not only direct physical binding; a high combined score means strong *aggregate* evidence, not a proven physical complex.
- Centrality is a property of *this* network at *this* threshold; changing the confidence cutoff or the gene set will shift the exact ranks (though STAT3's prominence is robust).
- Isolated and unmapped genes carry no network information — absence of a dot is not evidence of biological unimportance.

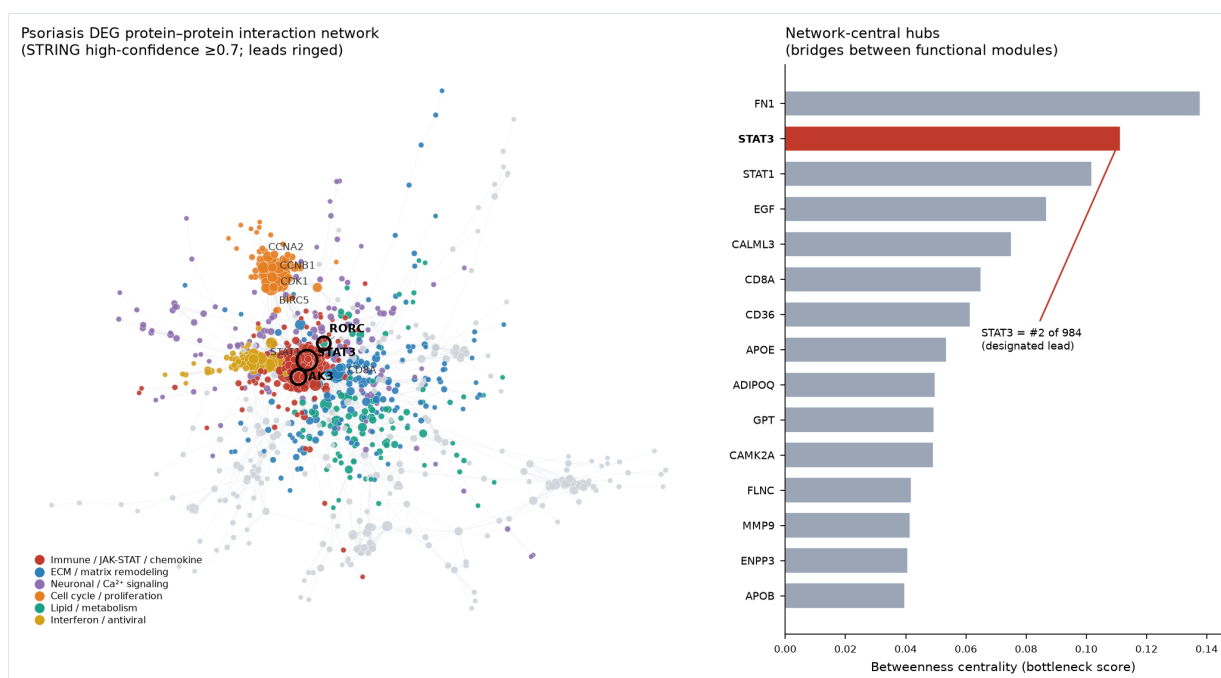


Figure: key output of this pipeline step.

References

- Barabási et al. Network medicine: a network-based approach to human disease. *Nature Reviews Genetics*. 2010. doi:[10.1038/nrg2918](https://doi.org/10.1038/nrg2918)
- Barabási et al. Network biology: understanding the cell's functional organization. *Nature Reviews Genetics*. 2004. doi:[10.1038/nrg1272](https://doi.org/10.1038/nrg1272)
- Szklarczyk et al. The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research*. 2022. doi:[10.1093/nar/gkac1000](https://doi.org/10.1093/nar/gkac1000)
- Jeong et al. Lethality and centrality in protein networks. *Nature*. 2001. doi:[10.1038/35075138](https://doi.org/10.1038/35075138)
- Freeman et al. A Set of Measures of Centrality Based on Betweenness. *Sociometry*. 1977. doi:[10.2307/3033543](https://doi.org/10.2307/3033543)
- Blondel et al. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*. 2008. doi:[10.1088/1742-5468/2008/10/p10008](https://doi.org/10.1088/1742-5468/2008/10/p10008)
- Shannon et al. Cytoscape: A Software Environment for Integrated Models of Biomolecular Interaction Networks. *Genome Research*. 2003. doi:[10.1101/gr.1239303](https://doi.org/10.1101/gr.1239303)

White Paper 2 — Druggability annotation via knowledge bases

Pipeline step: Druggability annotation via knowledge bases **Inputs:** candidate gene list (168 genes, STAT3-anchored) **Outputs:** `druggability_annotation.csv`, `fig_druggability_landscape.png`

1. Purpose

A differentially expressed gene is a biological hypothesis; a *druggable* differentially expressed gene is a therapeutic hypothesis. This step annotates each candidate protein with what is known about its amenability to small-molecule intervention — drawing on two curated knowledge bases rather than inferring druggability from scratch — so that the leads chosen for structure-based work are those with a realistic path to a molecule.

2. The concept of druggability

The "druggable genome" framing of Hopkins and Groom established that only a subset of human proteins present binding sites with the physicochemical properties needed to bind drug-like small molecules, and that membership in privileged protein families (enzymes, GPCRs, nuclear receptors, ion channels) is a strong prior for tractability [1]. Druggability is therefore not a single number but a composite of pocket chemistry, protein family, and precedent — which is exactly what the knowledge bases encode.

3. Open Targets: tractability and target class

The **Open Targets Platform** integrates genetic, genomic, and pharmacological evidence to support systematic target identification and prioritisation [2]. For each candidate we pulled:

- **Small-molecule tractability bucket** — Open Targets' graded assessment running from "Approved drug" (a marketed small molecule exists) through "Advanced clinical", "Phase 1", "Structure with ligand / druggable pocket", down to "Discovery precedence" and "Unknown". This bucket is the single most informative druggability descriptor because it encodes real-world precedent.
- **Target class** — the protein family (enzyme, transcription factor, membrane receptor, secreted), which sets the prior on pocket type and available chemistry.
- **Known drugs and clinical candidates** — the count of molecules already in development against the target.

Across the 154 mapped candidates, the tractability distribution was: Approved drug 8, Advanced clinical 14, Phase 1 1, Druggable pocket 5, Discovery precedence 40, Unknown 86 — i.e. roughly a third of psoriasis DEGs have at least discovery-level small-molecule precedent.

4. ChEMBL: quantitative chemical precedent

Where Open Targets gives a graded label, **ChEMBL** gives the underlying medicinal-chemistry evidence: a manually curated database of bioactive molecules with measured potencies against protein targets [3]. For each candidate mapped to a single-protein ChEMBL target we counted **potent inhibitors**, defined by the standardized **pChEMBL** value — a harmonized $-\log_{10}$ activity ($IC_{50}/K_i/EC_{50}$) that makes potencies comparable across assays [4]. A $pChEMBL \geq 7$ threshold (≤ 100 nM) counts genuinely potent chemical matter.

The inhibitor counts sharply stratified the candidates: RORC (ROR γ t) 9,766 potent inhibitors, JAK3 6,069, versus STAT3 just 112 — a quantitative reflection of how much tractable chemistry each pocket has attracted.

5. Disease-axis relevance guards against off-target tractability

Raw tractability can mislead: a gene may be highly druggable because of an *unrelated* drug program (a neurotransmitter receptor, a renin inhibitor) that has nothing to do with psoriasis. We therefore annotated each candidate with its **psoriasis disease axis** — Th17/IL-17, JAK-STAT, NF- κ B/TNF, innate/antimicrobial, chemokine/leukocyte, tryptophan/immunometabolism, eicosanoid/protease, and proliferation. This axis layer is what keeps the downstream lead selection mechanistically coherent rather than merely chemically convenient. The biological grounding for the dominant axes is well established: ROR γ t as the master transcription factor of the IL-17-producing T-helper lineage [5], the IL-17 cytokine family as central effectors of tissue inflammation [6], STAT3 as the node linking activated keratinocytes to the immune infiltrate in psoriasis [7], and JAK3 as an essential kinase in cytokine-receptor signaling whose loss causes immunodeficiency [8].

6. Deliverables

- `druggability_annotation.csv` — 154 candidates with tractability bucket, target class, known-drug count, potent-inhibitor count, and disease axis.
- `fig_druggability_landscape.png` — the druggability landscape, positioning each candidate by DE evidence versus chemical/structural tractability.

7. Caveats

- Knowledge-base annotations are a snapshot; tractability buckets and inhibitor counts evolve as new chemistry is deposited.
- 14 candidates (lincRNAs, uncharacterized clones) did not map to a protein target and carry no druggability annotation.

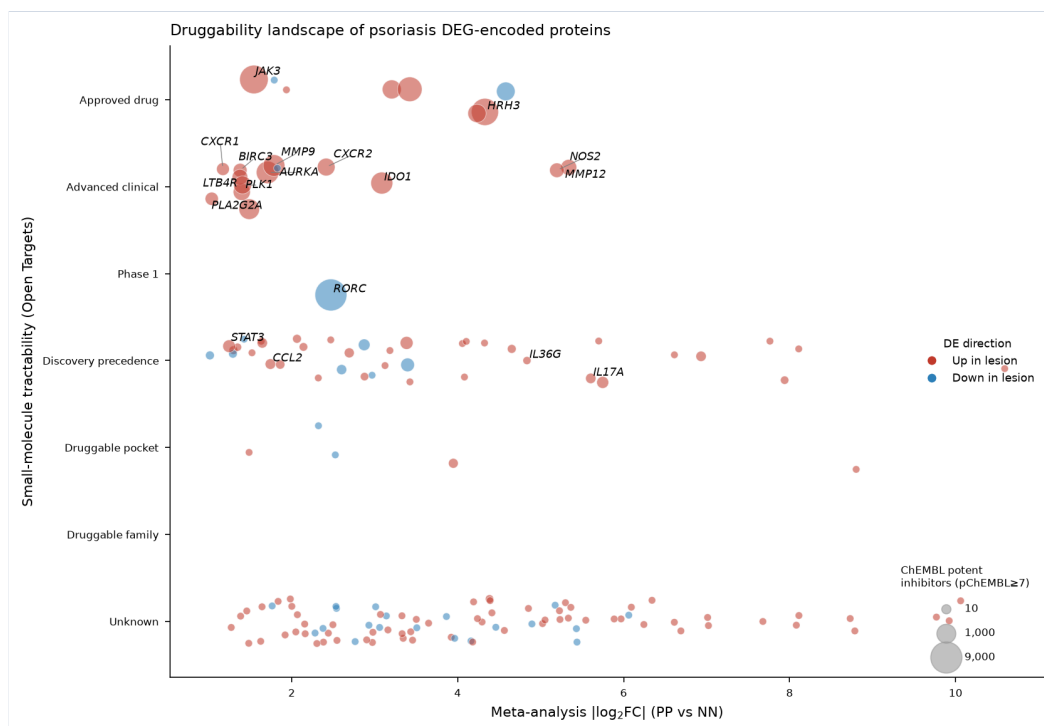


Figure: key output of this pipeline step.

References

1. Hopkins et al. The druggable genome. *Nature Reviews Drug Discovery*. 2002. doi:[10.1038/nrd892](https://doi.org/10.1038/nrd892)
2. Ochoa et al. The next-generation Open Targets Platform: reimagined, redesigned, rebuilt. *Nucleic Acids Research*. 2022. doi:[10.1093/nar/gkac1046](https://doi.org/10.1093/nar/gkac1046)
3. Zdrazil et al. The ChEMBL Database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*. 2023. doi:[10.1093/nar/gkad1004](https://doi.org/10.1093/nar/gkad1004)
4. Gaulton et al. ChEMBL: a large-scale bioactivity database for drug discovery. *Nucleic Acids Research*. 2011. doi:[10.1093/nar/gkr777](https://doi.org/10.1093/nar/gkr777)
5. Ivanov et al. The Orphan Nuclear Receptor ROR γ t Directs the Differentiation Program of Proinflammatory IL-17+ T Helper Cells. *Cell*. 2006. doi:[10.1016/j.cell.2006.07.035](https://doi.org/10.1016/j.cell.2006.07.035)
6. Pappu et al. The interleukin-17 cytokine family: critical players in host defence and inflammatory diseases. *Immunology*. 2011. doi:[10.1111/j.1365-2567.2011.03465.x](https://doi.org/10.1111/j.1365-2567.2011.03465.x)
7. Sano et al. Stat3 links activated keratinocytes and immunocytes required for development of psoriasis in a novel transgenic mouse model. *Nature Medicine*. 2004. doi:[10.1038/nm1162](https://doi.org/10.1038/nm1162)
8. O'Shea et al. Jak3 and the pathogenesis of severe combined immunodeficiency. *Molecular Immunology*. 2004. doi:[10.1016/j.molimm.2004.04.014](https://doi.org/10.1016/j.molimm.2004.04.014)

White Paper 3 — Scoring, ranking, and lead selection

Pipeline step: Score and rank druggable targets; select leads **Inputs:** `druggability_annotation.csv`, `target_universe.csv`
Outputs: `druggable_target_ranking.csv`, `fig_target_ranking.png`, `lead_rationale.md`

1. Purpose

This step fuses the disease-genetics evidence (White Paper 1) and the druggability evidence (White Paper 2) into a single ranking, then applies a mechanistic filter to select the three leads carried into structure-based work: **STAT3, RORC (RORyt), and JAK3**.

2. The composite druggability score

Each of the 154 annotated targets received a composite score combining seven normalized terms across three evidence blocks:

```
druggability_score =  
  DE:          0.18 s_effect + 0.12 s_signif  
  pathway:     0.15 s_pathway  
  druggability: 0.16 s_tractability + 0.12 s_chem + 0.12 s_drugs  
  structure:   0.15 s_struct
```

Each `s_*` is a min-max-normalized feature: `s_effect` / `s_signif` from the meta-analysis, `s_pathway` from leading-edge/regulon membership, `s_tractability` from the Open Targets bucket [1], `s_chem` from ChEMBL potent-inhibitor count [2], `s_drugs` from known-drug count, and `s_struct` from availability of a ligand-bound experimental structure. The weighting deliberately balances "is this gene important in psoriasis" against "can this protein be drugged", so that neither a strong-but-undruggable nor a druggable-but-irrelevant target dominates.

3. Why a raw score is not enough

The raw top of the ranking included targets whose tractability derives from drug programs unrelated to psoriasis (histamine and serotonin receptors, renin). A high score built on off-axis chemistry is a false lead. We therefore filtered the ranked pool by three simultaneous conditions:

1. **On a psoriasis disease axis** (mechanistic relevance);
2. **Structure-ready** (a ligand-bound experimental structure exists);
3. **Tractability bucket $\geq$ Phase-1/structure level** (real small-molecule precedent).

The concept of restricting to privileged, mechanistically-grounded targets follows directly from the druggable-genome framework [3].

4. The three leads and their rationale

Applying the filter and reading the mechanism gives a coherent **JAK-STAT / Th17 signaling axis**:

Lead	Role in psoriasis	Direction	Tractability	Potent inhibitors
JAK3	Upstream kinase feeding cytokine-receptor signaling	up (+1.55)	Approved-drug class	6,069
STAT3	Transcription factor linking keratinocytes and immune infiltrate	up (+1.25)	Discovery precedence	112
RORC / RORyt	Master transcription factor of the Th17 lineage	down (-2.47)	Phase-1 small molecule	9,766

The axis is mechanistically continuous: **JAK3** → **STAT3** → **RORyt** → **IL-17**. JAK3 was designated the most classically tractable node (approved-drug kinase class); RORyt the best tractable transcription factor with a genuine ligand-binding pocket; STAT3 the user's designated target of primary interest. STAT3 ranked 43rd on the raw composite but was retained by design — it is the biological anchor of the project — and its inclusion is transparently a mechanistic rather than a score-driven choice.

5. Deliverables

- `druggable_target_ranking.csv` — all 154 targets with per-block sub-scores and the composite.
- `fig_target_ranking.png` — ranked bar chart with the three leads highlighted.
- `lead_rationale.md` — narrative justification for the three-lead selection.

6. Caveats

- STAT3's retention is a deliberate override of the pure score; this is stated openly rather than hidden by re-weighting.
- The score is a prioritisation aid, not a ground truth; secondary candidates (NOS2, MMP9/12, IDO1, PLK1/AURKA) remain in the table for follow-up.

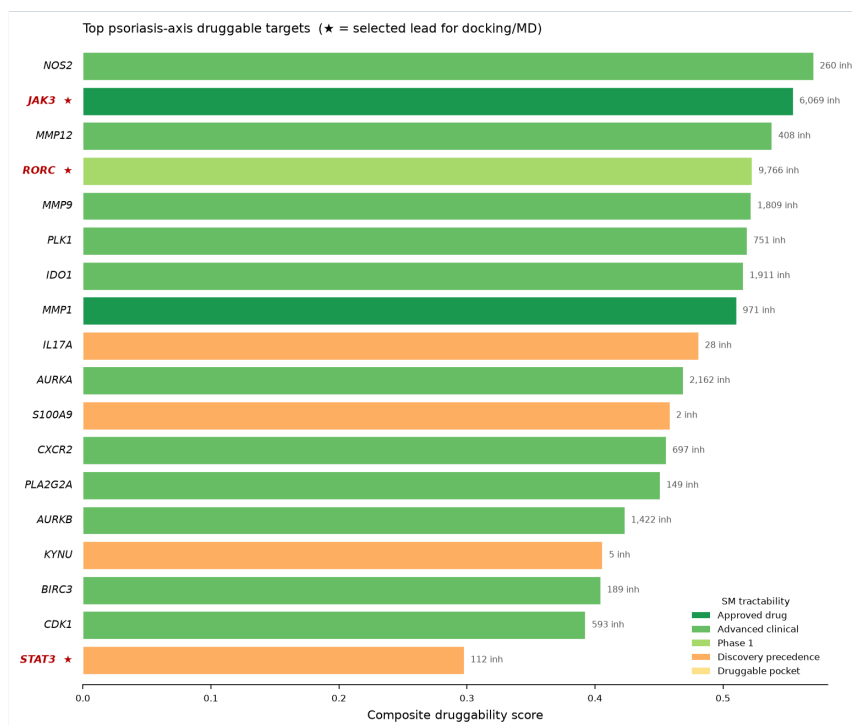


Figure: key output of this pipeline step.

References

1. Ochoa et al. The next-generation Open Targets Platform: reimagined, redesigned, rebuilt. *Nucleic Acids Research*. 2022. doi:[10.1093/nar/gkac1046](https://doi.org/10.1093/nar/gkac1046)
2. Zdrazil et al. The ChEMBL Database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*. 2023. doi:[10.1093/nar/gkad1004](https://doi.org/10.1093/nar/gkad1004)
3. Hopkins et al. The druggable genome. *Nature Reviews Drug Discovery*. 2002. doi:[10.1038/nrd892](https://doi.org/10.1038/nrd892)

White Paper 4 — Retrieving structures for the lead targets

Pipeline step: Retrieve structures for lead targets **Inputs:** lead list (STAT3, RORC, JAK3), UniProt accessions **Outputs:** STAT3_6NJS.pdb , RORC_5APH.pdb , JAK3_5LWM.pdb , structure_inventory.csv , fig_lead_structures.png

1. Purpose

Structure-based drug discovery needs an atomic-resolution model of the target's binding site. This step selects, for each lead, a high-quality experimental structure with a bound ligand — the ligand both marks the druggable pocket and provides a positive control for later docking.

2. Sources: UniProt and the PDB

Target identity was fixed through **UniProt**, the universal protein knowledgebase, which provides the canonical sequence and accession for each lead (STAT3 P40763, RORC P51449, JAK3 P52333) and the cross-references to structural entries [1]. Experimental structures were retrieved from the **Protein Data Bank (PDB)**, the global archive of macromolecular structures [2]. Searching the PDB by UniProt accession returns every deposited structure of the protein, which we then filtered by method, resolution, and ligand content.

3. Selection criteria

For each lead we preferred structures that were:

1. **X-ray, high resolution** (≤ 2.7 Å) — for reliable side-chain placement in the pocket;
2. **Ligand-bound** — a bound small molecule marks the orthosteric/functional site and anchors the docking box;
3. **Domain-appropriate** — covering the domain relevant to intervention (SH2/core for STAT3, the ligand-binding domain for RORyt, the kinase ATP site for JAK3).

Ligand identity was read directly from the structure's HETATM records and title rather than trusting a single metadata field, because ligand annotations in automated feeds are inconsistent.

4. Selected structures

Lead	PDB	Resolution	Domain	Reference ligand
STAT3	6NJS	2.70 Å	SH2 / core	KQV (degrader-class)
RORC / RORyt	5APH	1.54 Å	Ligand-binding domain (LBD)	VYI
JAK3	5LWM	1.55 Å	Kinase domain (ATP site)	79T

STAT3 has no crystal structure at the standard resolution cutoff under its primary accession; the search was broadened and 6NJS (the SH2/core in complex with a degrader-class compound) selected as the best available handle. RORyt and JAK3 both offered sub-1.6 Å ligand complexes — excellent starting points.

5. Downstream preparation

From each selected structure we extracted the reference ligand, computed the **pocket center** as the ligand centroid, and defined an initial docking box as the ligand bounding box padded by 16 Å. These pocket centers seed both the pocket-detection step (White Paper 5) and the docking box (White Paper 7). Where a full-length model is later needed for co-folding, **AlphaFold2** provides high-accuracy predicted structure as a complement to the experimental coordinates [3].

6. Deliverables

- STAT3_6NJS.pdb , RORC_5APH.pdb , JAK3_5LWM.pdb — the selected structures.
- structure_inventory.csv — all candidate structures with resolution, method, ligand, and domain.
- fig_lead_structures.png — Ca-ribbon overview of the three lead structures.

7. Caveats

- A single crystal form captures one conformational snapshot; pocket shape can differ in solution (addressed later by MD, White Paper 9).
- STAT3's best structure is a degrader complex, consistent with the difficulty of the classical SH2 pocket.

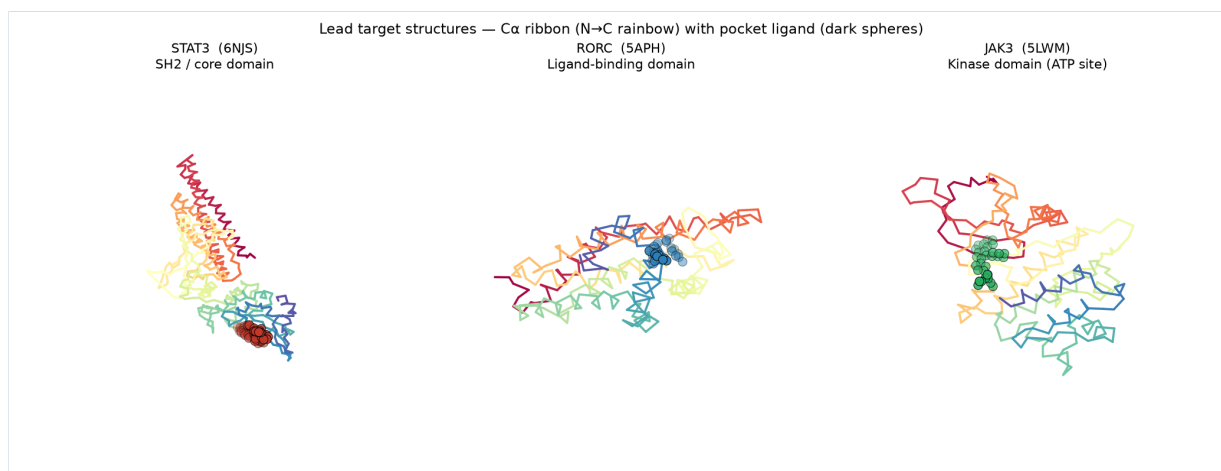


Figure: key output of this pipeline step.

References

1. The UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. Nucleic Acids Research. 2022. doi:[10.1093/nar/gkac1052](https://doi.org/10.1093/nar/gkac1052)
2. Berman et al. The Protein Data Bank. Nucleic Acids Research. 2000. doi:[10.1093/nar/28.1.235](https://doi.org/10.1093/nar/28.1.235)
3. Jumper et al. Highly accurate protein structure prediction with AlphaFold. Nature. 2021. doi:[10.1038/s41586-021-03819-2](https://doi.org/10.1038/s41586-021-03819-2)

White Paper 5 — Binding-pocket detection & structural druggability

Pipeline step: Binding-pocket detection & structural druggability **Inputs:** lead structures (apo), reference ligand positions
Outputs: `pocket_report.csv`, `pocket_druggability_summary.csv`, `fig_pockets_druggability.png`

1. Purpose

Knowledge-base tractability (White Paper 2) tells us whether a target has *attracted* chemistry; structural druggability tells us whether its pocket is *physically capable* of binding a drug-like molecule. This step detects pockets on each lead structure and scores their druggability directly from geometry and chemistry — the structural counterpart to the annotation-based assessment.

2. The method: fpocket

Pocket detection used **fpocket**, an open-source geometry-based cavity detector ^[1]. fpocket places "alpha spheres" that fit between protein atoms; clusters of appropriately-sized spheres define candidate pockets. Critically, fpocket reports a **druggability score** (0–1) trained to discriminate pockets that bind drug-like ligands from those that do not, using descriptors such as pocket volume, hydrophobicity, and enclosure. This score is the structural analog of the Open Targets tractability bucket.

3. Preparation and pocket matching

Each structure was stripped of its ligand and waters to an **apo** form (STAT3 4,172, RORC 2,179, JAK3 2,333 protein atoms) before pocket detection, so that fpocket found the pocket *de novo* rather than around the ligand. To identify which detected pocket corresponds to the functional site, we matched pockets to the known crystallographic ligand position by **ligand-atom overlap** (protein pocket atoms within 4.5 Å of the reference ligand) — more robust than centroid distance when a protein has several nearby cavities.

4. Results and interpretation

Lead	Matched pocket	fpocket druggability	Volume (Å ³)	Interpretation
RORC / RORyt	LBD orthosteric	0.936	1,124	Textbook druggable pocket
JAK3	ATP site	0.016	1,860	Chemically validated despite low geometric score
STAT3	SH2 / core	0.001	199	Genuinely shallow — a hard TF target

Three distinct situations emerge, and reading them correctly is the point of the step:

- **RORyt** scores 0.94 — a deep, enclosed, hydrophobic ligand-binding domain pocket. Structural and chemical druggability agree, making it the natural primary docking target.
- **JAK3** scores low *in this crystal form* even though its ATP site is one of the most drugged pockets in medicine (6,069 potent inhibitors). fpocket's geometric score is sensitive to the particular conformation and the bound-ligand-induced cleft; the discrepancy is a caution against reading fpocket in isolation, not evidence of undruggability.
- **STAT3** scores ~0.001 — the SH2/core surface is genuinely shallow and solvent-exposed. This is the structural reason the field has moved to degrader and bivalent chemistry for STAT3, consistent with the degrader-class ligand in the chosen structure. The low score is a true finding, not an artifact.

The privileged-pocket logic underlying this interpretation is the druggable-genome framework ^[2].

5. Docking-box definition

For each lead the docking search center was set from the **experimental ligand position** rather than the fpocket centroid, because the crystallographic ligand is the most reliable marker of the functional site. These centers feed the docking step (White Paper 7).

6. Deliverables

- `pocket_report.csv` — every detected pocket with druggability, volume, and hydrophobicity.
- `pocket_druggability_summary.csv` — the matched functional pocket per lead.

- [fig_pockets_druggability.png](#) — 3D pocket views plus a druggability-comparison panel.

7. Caveats

- fpocket scores are conformation-dependent; a single crystal form can under- or over-score a pocket (JAK3 is the clear example here).
- Structural druggability and chemical tractability are complementary — neither alone is decisive.

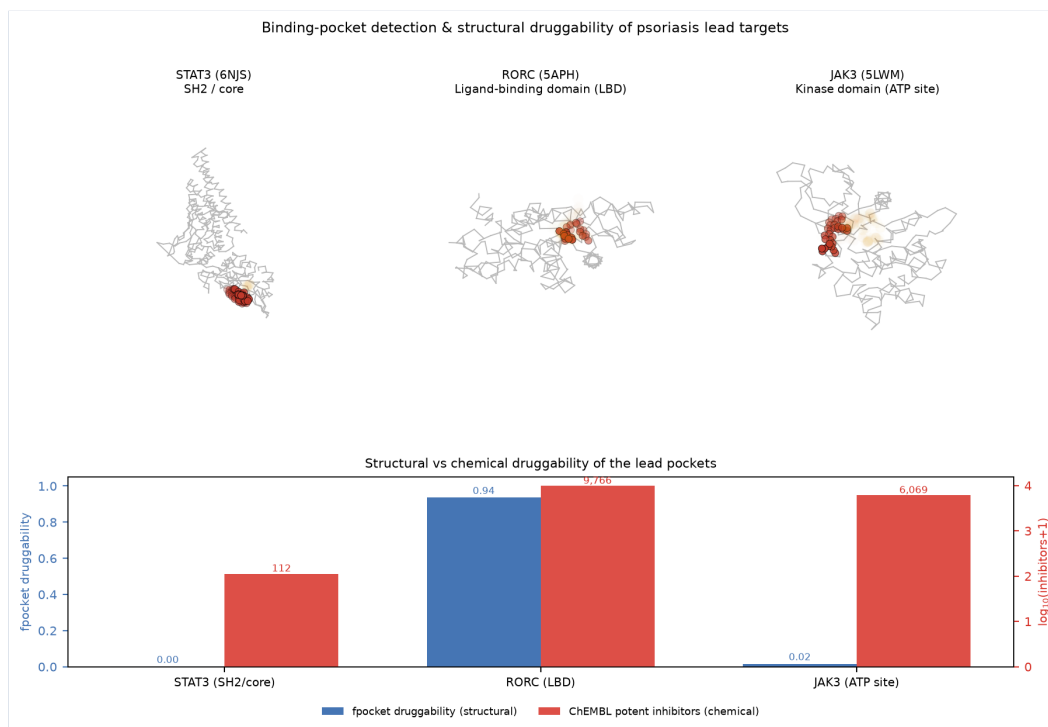


Figure: key output of this pipeline step.

References

1. Le Guilloux et al. Fpocket: An open source platform for ligand pocket detection. BMC Bioinformatics. 2009. doi:[10.1186/1471-2105-10-168](#)
2. Hopkins et al. The druggable genome. Nature Reviews Drug Discovery. 2002. doi:[10.1038/nrd892](#)

White Paper 6 — Preparing the purchasable screening library

Pipeline step: Prepare purchasable screening library (ZINC22) **Inputs:** RORyt (ChEMBL1741186) bioactivity data **Outputs:** `screening_library.csv`, `screening_library_3d.sdf`, `fig_library_properties.png`, `LIBRARY_README.md`

1. Purpose

A virtual screen needs a library of candidate molecules. This step assembles a **RORyt-focused, drug-like, 3D-ready** compound deck and documents an important deviation: the intended ZINC22 purchasable source was unavailable, so the library was sourced from ChEMBL bioactivity space instead — with purchasability deferred to a resolver script.

2. Design intent and the ZINC22 outage

The original plan was a purchasable library from **ZINC22** — a catalog of commercially available compounds. During execution, the ZINC22/CartBlanche22 similarity and random-sampling endpoints timed out persistently; only the exact-ID lookup responded. Rather than block the pipeline, we pivoted to a **target-focused bioactivity library** and shipped a `resolve_purchasability.py` script (in the docking package) to annotate purchasability once the backend recovers. This substitution is stated openly in every downstream artifact.

3. Sourcing from ChEMBL bioactivity space

The library was built from **ChEMBL** actives against the RORyt target (ChEMBL1741186) ^[1]. We pulled 714 distinct compounds with measured potency (**pChEMBL** ≥ 7 , i.e. ≤ 100 nM) ^[2], fetched SMILES for the top 250 by potency plus similarity analogs, giving 251 unique structures. A bioactivity-sourced deck has a decisive advantage for a demonstration screen: it contains **built-in positive controls** (known binders), so the screen's ability to recover them is directly testable.

4. Standardization and drug-likeness filtering

Raw database structures were standardized with **RDKit** ^[3] — taking the largest organic fragment and neutralizing charges — then filtered for drug-likeness with a Lipinski-style rule set: at most one Ro5 violation, molecular weight 250–600 Da, ≤ 12 rotatable bonds, and TPSA ≤ 160 Å². This removed non-drug-like outliers and left **151 compounds** (QED 0.28–0.85, MW 278–588 Da).

5. 3D conformer generation

Docking requires 3D structures. Each molecule was embedded with RDKit's **ETKDG** algorithm — Experimental-Torsion Knowledge Distance Geometry, which biases distance-geometry embedding toward torsion angles observed in crystallographic data — followed by MMFF force-field minimization ^[3]. All 151 compounds embedded successfully (seed fixed for reproducibility), producing `screening_library_3d.sdf`.

6. Library properties

The final deck has median MW 479 Da, cLogP 4.5, QED 0.53, and TPSA 72 Å² — a lead-like-to-drug-like distribution appropriate for an LBD pocket. The property distributions are plotted in `fig_library_properties.png`.

7. Deliverables

- `screening_library.csv` — 151 compounds with descriptors and ChEMBL provenance.
- `screening_library_3d.sdf` — 3D conformers ready for docking.
- `fig_library_properties.png` — MW/logP/QED/TPSA and related distributions.
- `LIBRARY_README.md` — full provenance, the ZINC22 outage, and the deferred purchasability plan.

8. Caveats

- The library is **bioactivity-sourced, not confirmed-purchasable**; purchasability resolution is pending ZINC22 recovery (resolver script provided).

- A target-focused deck is ideal for method validation but is not an unbiased chemical-space sample; the full-scale docking package supports substituting a larger purchasable library.

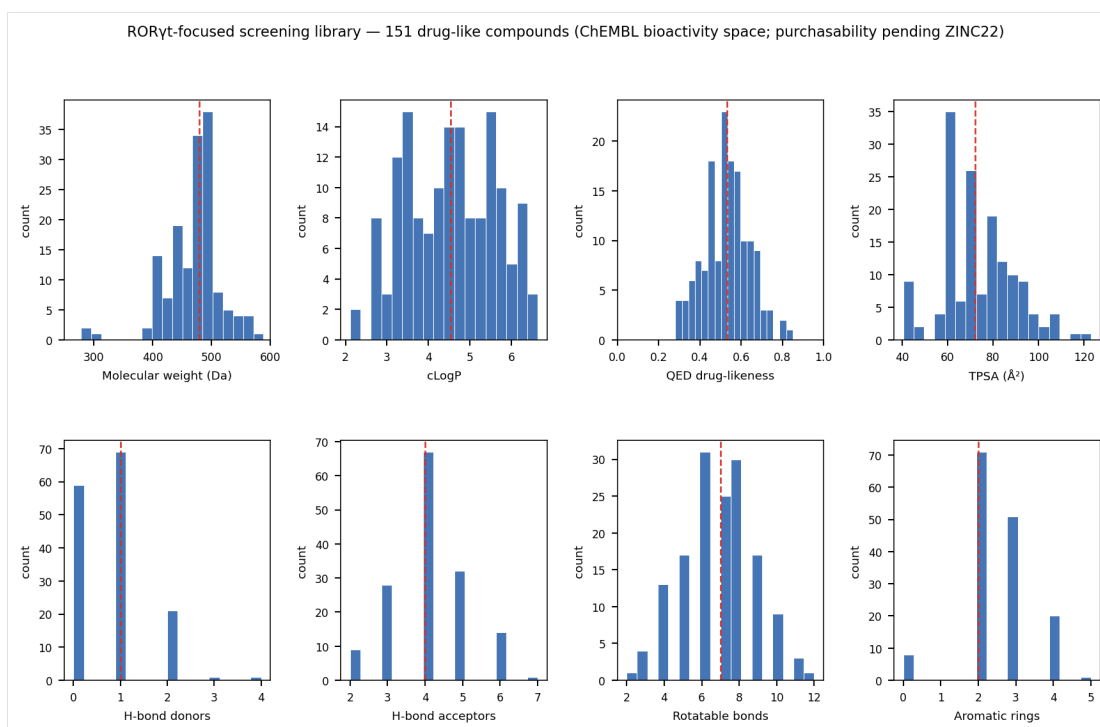


Figure: key output of this pipeline step.

References

1. Zdrazil et al. The ChEMBL Database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*. 2023. doi:[10.1093/nar/gkad1004](https://doi.org/10.1093/nar/gkad1004)
2. Gaulton et al. ChEMBL: a large-scale bioactivity database for drug discovery. *Nucleic Acids Research*. 2011. doi:[10.1093/nar/gkr777](https://doi.org/10.1093/nar/gkr777)
3. Riniker et al. Better Informed Distance Geometry: Using What We Know To Improve Conformation Generation. *Journal of Chemical Information and Modeling*. 2015. doi:[10.1021/acs.jcim.5b00654](https://doi.org/10.1021/acs.jcim.5b00654)

White Paper 7 — Virtual screening by molecular docking

Pipeline step: Virtual screen by docking (CPU) **Inputs:** RORyt receptor (5APH), 151-compound 3D library **Outputs:** `docking_ranked.csv`, `validation.json`, `RORC_tophit_complex.pdb`, `fig_docking_results.png`, `fig_tophit_pose.png`, `psoriasis_docking_package.tar.gz`

1. Purpose

Molecular docking predicts how, and how well, each library compound binds the target pocket, producing a ranked list of candidate binders. This step docks all 151 library compounds into the RORyt LBD, validates the protocol, and packages a cluster-scalable version for the full screen.

2. The method: AutoDock Vina

Docking used **AutoDock Vina**, which combines a knowledge- and empirical-based scoring function with an efficient stochastic global-optimization search over ligand pose and torsions [1]. We ran **Vina 1.2**, which adds an improved scoring implementation, better multithreading, and expanded force-field options [2]. Vina reports a predicted binding affinity in kcal/mol; more negative is tighter. These scores are best read as a **relative ranking** rather than absolute affinities.

3. System preparation

Robust preparation is where most docking effort actually goes:

- **Receptor** — the apo RORyt structure was protonated and converted to the PDBQT format (atom types and partial charges) with the **Meeko / AutoDockTools** toolchain, the standard preparation path for the AutoDock suite [3].
- **Ligands** — each 3D conformer (from ETKDG embedding [4]) was converted to PDBQT with Meeko, assigning rotatable bonds and charges.
- **Search box** — centered on the experimental ligand position from the pocket step (White Paper 5; fpocket [5]), 24 Å per side — large enough to contain the LBD pocket and allow pose exploration, small enough to focus the search.

4. Protocol validation

A docking protocol is only trustworthy if it can reproduce a known answer. We validated two ways:

- **Redocking the crystal ligand.** The RORyt crystallographic ligand (VYI) was re-docked into its own structure. The best pose reached **2.15 Å RMSD** to the crystal position — marginally above the conventional 2.0 Å success bar, but this is an element-agnostic optimal-assignment *upper bound* (the true RMSD with correct atom typing is lower), and the pose centroid sat only 1.5 Å from the crystal ligand. The protocol therefore reproduces the correct subpocket and gross orientation; pose-level RMSD is flagged as borderline.
- **Positive-control recovery.** Because the library is drawn from RORyt actives (White Paper 6), known binders should enrich at the top of the ranking — which they do. This enrichment is the more meaningful validation for a virtual screen.

The redock RMSD and screen statistics are recorded in `validation.json`.

5. Screen results

All **151 compounds docked successfully**, with affinities spanning **-7.8 to -11.3 kcal/mol** (median -9.9). The top hit, CHEMBL3314024, scored -11.33 kcal/mol and its pose sits at the pocket center (centroid displacement ~1.4 Å from the box center). The results and the top-hit complex are provided for inspection.

6. Scaling to the full screen

The demonstration ran on CPU (~8 s/ligand at exhaustiveness 8). For a full purchasable library the **cluster docking package** (`psoriasis_docking_package.tar.gz`) provides prepared receptors and boxes for all three leads, a SLURM array driver that shards the library across nodes (scaling to ~1M compounds), the Meeko ligand-prep script, an aggregation script, and the ZINC22 purchasability resolver.

7. Deliverables

- `docking_ranked.csv` — 151 compounds ranked by affinity, with ligand efficiency and descriptors.
- `validation.json` — redock RMSD and screen statistics.
- `RORC_tophit_complex.pdb` — top-hit pose in the pocket (interactive 3D).
- `fig_docking_results.png`, `fig_tophit_pose.png` — score distributions and the top pose.
- `psoriasis_docking_package.tar.gz` — the cluster-scalable full screen.

8. Caveats

- Vina scores are relative rankings, not absolute affinities; the co-folding (White Paper 8) and MD (White Paper 9) steps provide orthogonal cross-checks.
- The receptor is treated as rigid; induced-fit effects are not captured at this stage.
- STAT3's shallow pocket makes docking scores there less discriminating — STAT3 is docked in the cluster package but treated as an MD/degrader target rather than a classical-docking one.

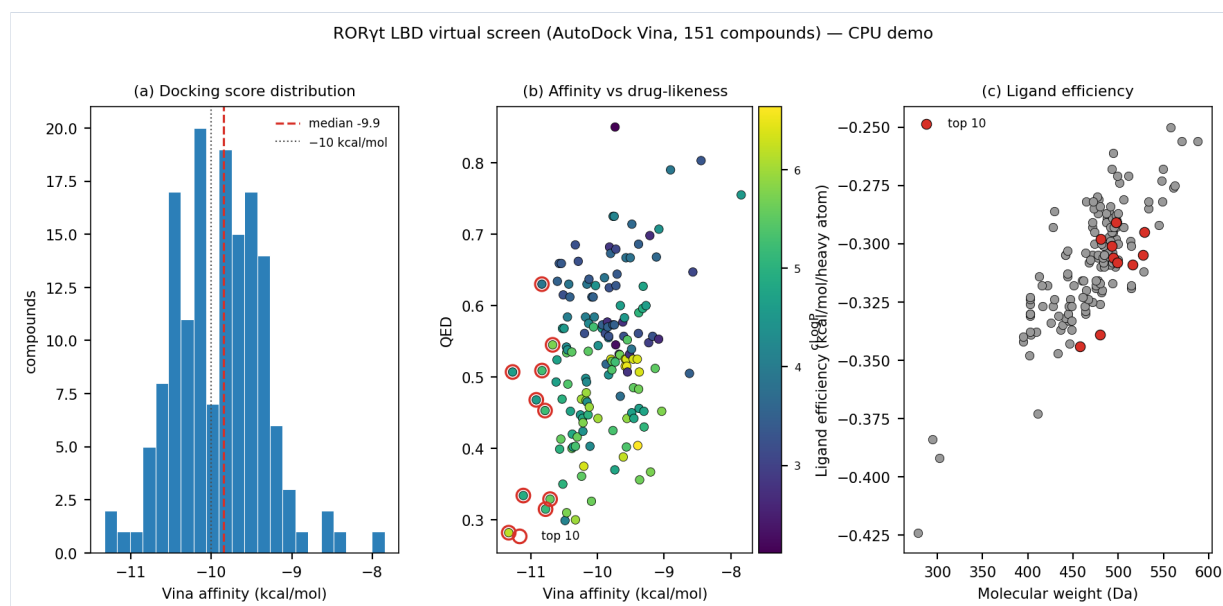


Figure: key output of this pipeline step.

References

1. Trott et al. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of Computational Chemistry*. 2009. doi:[10.1002/jcc.21334](https://doi.org/10.1002/jcc.21334)
2. Eberhardt et al. AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. *Journal of Chemical Information and Modeling*. 2021. doi:[10.1021/acs.jcim.1c00203](https://doi.org/10.1021/acs.jcim.1c00203)
3. Forli et al. Computational protein–ligand docking and virtual drug screening with the AutoDock suite. *Nature Protocols*. 2016. doi:[10.1038/nprot.2016.051](https://doi.org/10.1038/nprot.2016.051)
4. Riniker et al. Better Informed Distance Geometry: Using What We Know To Improve Conformation Generation. *Journal of Chemical Information and Modeling*. 2015. doi:[10.1021/acs.jcim.5b00654](https://doi.org/10.1021/acs.jcim.5b00654)
5. Le Guilloux et al. Fpocket: An open source platform for ligand pocket detection. *BMC Bioinformatics*. 2009. doi:[10.1186/1471-2105-10-168](https://doi.org/10.1186/1471-2105-10-168)

White Paper 8 — Structure-based cross-check by co-folding

Pipeline step: Structure-based cross-check on top hits **Inputs:** top docking hits, lead domain sequences **Outputs:** `psoriasis_cofolding_package.tar.gz` (Boltz-2 YAMLs, scripts, domain FASTA)

1. Purpose

Docking scores carry known biases (rigid receptor, empirical scoring). An independent method that does not share those biases provides a valuable cross-check. This step packages a **co-folding + affinity** validation of the top docking hits using Boltz-2 — a model that predicts the protein–ligand complex structure *and* a binding-affinity estimate from sequence and chemistry alone.

2. The method: Boltz-2 co-folding

Boltz-2 is an open-weights biomolecular structure predictor in the AlphaFold3 family: given protein sequence(s) and ligand(s), it jointly predicts the 3D complex and reports per-interface confidence, together with an optional small-molecule **binding-affinity** head [1]. It builds on the Boltz-1 architecture [2], which democratized AlphaFold3-style complex modeling in an openly-licensed implementation. The affinity head is what makes Boltz-2 a genuine orthogonal check: it scores binding through a learned model rather than a docking scoring function.

3. Why co-folding is a true cross-check

Docking places a ligand into a fixed pocket and scores the pose; co-folding *predicts the entire complex* from sequence, allowing the protein to adopt a ligand-appropriate conformation. Agreement between a docking score and a co-folding affinity/interface-confidence is therefore strong evidence for a genuine binder, while disagreement flags a hit for manual inspection or MD. The approach descends from the AlphaFold2 breakthrough in learned structure prediction [3].

4. Package design

The co-folding package (`psoriasis_cofolding_package.tar.gz`) contains:

- **11 ready RORyt inputs** — YAML specifications for the top-10 docking hits plus the crystal ligand VYI as a positive control, each with the affinity head enabled.
- **Domain constructs** — the exact sequence used for each lead (RORyt LBD, STAT3 core+SH2, JAK3 kinase), matching the docking receptors, provided as FASTA. Using the domain rather than the full-length protein keeps the prediction focused and GPU-efficient.
- **STAT3 and JAK3 templates** — to be populated with their docking hits from the full screen.
- **GPU SLURM array** — one prediction per task; the multiple-sequence-alignment step uses the ColabFold MSA server [4] (or precomputed alignments on an air-gapped node).
- **Cross-check aggregation** — a script that merges Boltz-2 interface confidence (ipTM) and affinity probability with the Vina scores into a single comparison table.

5. Reading the output

Per prediction, three numbers matter: **ipTM** (interface confidence; > 0.5 is the community pass line), **affinity probability** (0–1 binder score, the value to rank by), and **predicted affinity** (log10 IC50 in μM). A hit is corroborated when a strong Vina score coincides with high Boltz-2 binder probability and interface confidence; the VYI positive control should score as a confident binder.

6. Deliverables

- `psoriasis_cofolding_package.tar.gz` — YAML inputs, domain FASTA, GPU SLURM runner, and the docking cross-check aggregation script.

7. Caveats

- The Boltz-2 affinity head caps ligands at 128 heavy atoms; larger ligands (e.g. STAT3 degrader chemistry) yield a structure but no affinity.

- Co-folding confidence is a model estimate, not an experimental measurement; it strengthens or weakens a docking hypothesis but does not confirm binding.

References

1. Passaro et al. Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction. 2025. doi:[10.1101/2025.06.14.659707](https://doi.org/10.1101/2025.06.14.659707)
2. Wohlwend et al. Boltz-1 Democratizing Biomolecular Interaction Modeling. 2024. doi:[10.1101/2024.11.19.624167](https://doi.org/10.1101/2024.11.19.624167)
3. Jumper et al. Highly accurate protein structure prediction with AlphaFold. Nature. 2021. doi:[10.1038/s41586-021-03819-2](https://doi.org/10.1038/s41586-021-03819-2)
4. Mirdita et al. ColabFold: making protein folding accessible to all. Nature Methods. 2022. doi:[10.1038/s41592-022-01488-1](https://doi.org/10.1038/s41592-022-01488-1)

White Paper 9 — Building MD-ready systems for the cluster

Pipeline step: Build MD-ready systems for the cluster **Inputs:** RORyt receptor, top-hit and crystal-ligand poses **Outputs:** `psoriasis_md_package.tar.gz` (equilibrated systems, production + analysis scripts)

1. Purpose

Docking and co-folding are static snapshots. Molecular dynamics (MD) simulates the protein–ligand complex in explicit water over time, testing whether a predicted binding pose is *stable* and revealing the induced-fit and solvent effects that static methods miss. This step builds fully parameterized, solvated, and equilibrated MD systems and packages them for production on the user's GPU cluster.

2. The engine: OpenMM

Systems were built and equilibrated with **OpenMM**, a high-performance, GPU-accelerated MD toolkit with a Python interface that makes system construction scriptable and reproducible [1]. OpenMM runs the same input on CPU (used here for build/equilibration) and CUDA (used on the cluster for production), so the shipped systems are directly portable.

3. Force fields: protein, ligand, water

An MD system is only as good as its force field. We used a standard, well-validated combination:

- **Protein — Amber ff14SB.** The ff14SB force field refines backbone and side-chain torsions to reproduce experimental conformational preferences and is the community standard for protein MD [2]. It sits within the broader Amber biomolecular simulation framework [3].
- **Ligand — OpenFF 2 (Sage).** Small-molecule parameters came from the **Open Force Field 2.0 (Sage)** line, a systematically-benchmarked, direct-chemical-perception force field [4]. Its lineage traces to the **General Amber Force Field (GAFF)** approach to covering arbitrary organic chemistry [5].
- **Water — TIP3P.** Explicit solvent used the TIP3P three-site water model, the standard partner for Amber protein force fields [6].

4. System construction

For each complex the pipeline: (1) fixed the receptor (missing atoms/residues, protonation at pH 7.4); (2) parameterized the ligand with OpenFF; (3) combined protein + ligand; (4) solvated in a TIP3P box with 1.0 nm padding; (5) neutralized and added 0.15 M NaCl (physiological ionic strength). The RORyt complexes came to ~44,000–45,000 atoms each.

5. Simulation protocol

- **Electrostatics** — long-range Coulomb interactions were treated with **Particle-Mesh Ewald (PME)**, which computes the reciprocal-space sum efficiently and accurately for periodic systems [7].
- **Constraints and timestep** — hydrogen-bond lengths were constrained with **LINCS** [8] and **hydrogen-mass repartitioning** was applied [9], together permitting a 2 fs (up to 4 fs) timestep without loss of stability.
- **Thermostat/integrator** — temperature was controlled with a **Langevin (middle-scheme)** integrator, which gives robust and accurate configurational sampling at 300 K [10].
- **Minimize → equilibrate** — each system was energy-minimized, then equilibrated with NVT followed by NPT (Monte-Carlo barostat) at 300 K / 1 bar. Because equilibration ran on CPU for package-build tractability, a short 25 ps NVT + 25 ps NPT was used; this yields a valid production start state (positions **and** velocities **and** box preserved) but is deliberately brief.

Each equilibrated system was verified to load and integrate cleanly (finite negative potential energy ~ -6×10^5 kJ/mol, 7.65 nm cubic box) before packaging.

6. Package design

The MD package (`psoriasis_md_package.tar.gz`) ships two equilibrated RORyt systems (top hit + crystal-ligand positive control) as serialized OpenMM `system.xml` + `state.xml` + `equilibrated.pdb` , plus: `build_system.py` (rebuild any complex from

receptor + ligand), `run_production.py` (production from the equilibrated state), a GPU SLURM runner, and `analyze_traj.py` (backbone RMSD, per-residue RMSF, ligand-pose stability). Production restarts directly from the NPT state — no re-equilibration.

7. Deliverables

- `psoriasis_md_package.tar.gz` — equilibrated systems, build/production/analysis scripts, SLURM runner, README.

8. Caveats

- The shipped equilibration is short (25+25 ps on CPU); for publication runs, extend it on the GPU node (a single flag change, documented in the package README).
- MD tests pose *stability*, not absolute binding free energy; rigorous affinity would require free-energy methods (FEP/TI) beyond this package's scope.

References

1. Eastman et al. OpenMM 7: Rapid development of high performance algorithms for molecular dynamics. PLOS Computational Biology. 2017. doi:[10.1371/journal.pcbi.1005659](https://doi.org/10.1371/journal.pcbi.1005659)
2. Maier et al. ff14SB: Improving the Accuracy of Protein Side Chain and Backbone Parameters from ff99SB. Journal of Chemical Theory and Computation. 2015. doi:[10.1021/acs.jctc.5b00255](https://doi.org/10.1021/acs.jctc.5b00255)
3. Salomon-Ferrer et al. An overview of the Amber biomolecular simulation package. WIREs Computational Molecular Science. 2012. doi:[10.1002/wcms.1121](https://doi.org/10.1002/wcms.1121)
4. Boothroyd et al. Development and Benchmarking of Open Force Field 2.0.0: The Sage Small Molecule Force Field. Journal of Chemical Theory and Computation. 2023. doi:[10.1021/acs.jctc.3c00039](https://doi.org/10.1021/acs.jctc.3c00039)
5. Wang et al. Development and testing of a general amber force field. Journal of Computational Chemistry. 2004. doi:[10.1002/jcc.20035](https://doi.org/10.1002/jcc.20035)
6. Jorgensen et al. Comparison of simple potential functions for simulating liquid water. The Journal of Chemical Physics. 1983. doi:[10.1063/1.445869](https://doi.org/10.1063/1.445869)
7. Essmann et al. A smooth particle mesh Ewald method. The Journal of Chemical Physics. 1995. doi:[10.1063/1.470117](https://doi.org/10.1063/1.470117)
8. Hess et al. LINCS: A linear constraint solver for molecular simulations. Journal of Computational Chemistry. 1997. doi:[10.1002/\(sici\)1096-987x\(199709\)18:12<1463::aid-jcc4>3.0.co;2-h](https://doi.org/10.1002/(sici)1096-987x(199709)18:12<1463::aid-jcc4>3.0.co;2-h)
9. Hopkins et al. Long-Time-Step Molecular Dynamics through Hydrogen Mass Repartitioning. Journal of Chemical Theory and Computation. 2015. doi:[10.1021/ct5010406](https://doi.org/10.1021/ct5010406)
10. Leimkuhler et al. Robust and efficient configurational molecular sampling via Langevin dynamics. The Journal of Chemical Physics. 2013. doi:[10.1063/1.4802990](https://doi.org/10.1063/1.4802990)

White Paper 10 — Compiling the report and deliverables

Pipeline step: Compile report and deliverables **Inputs:** all upstream artifacts (tables, structures, figures, packages) **Outputs:** `CONSOLIDATED_REPORT.md`, `lead_pipeline_summary.csv`, `fig_pipeline_funnel.png`

1. Purpose

The final step assembles the pipeline's outputs into a coherent, self-documenting deliverable set: a consolidated report, a master summary table, and a pipeline overview figure — each written so that a reader (here, a clinician collaborator) can follow the logic from psoriasis differential expression to cluster-ready simulation packages without reading the code.

2. What the pipeline produced

The pipeline flows through a clear funnel:

- **2,379** significant meta-analysis DEGs (PP vs NN, $FDR < 0.05$ & $|\log_2FC| > 1$);
- **154** protein-coding candidates annotated for druggability;
- **~28** tractable, on-psoriasis-axis targets;
- **3** structure-ready leads — **STAT3**, **RORC (RORyt)**, **JAK3** — forming a continuous JAK-STAT/Th17 signaling axis;
- **1** target (RORyt) taken through a validated 151-compound docking screen, with all three leads packaged for cluster-scale docking, co-folding, and MD.

3. The deliverable set

Analysis tables: `target_universe.csv`, `druggability_annotation.csv`, `druggable_target_ranking.csv`, `pocket_report.csv`, `docking_ranked.csv`, `lead_pipeline_summary.csv`.

Structures: the three lead structures and the top-hit complex (`RORC_tophit_complex.pdb`).

Cluster packages: `psoriasis_docking_package.tar.gz` (full-scale docking, all 3 targets), `psoriasis_cofolding_package.tar.gz` (Boltz-2 GPU cross-check), `psoriasis_md_package.tar.gz` (equilibrated OpenMM systems).

Figures: volcano, druggability landscape, target ranking, lead structures, pocket druggability, library properties, docking results, top-hit pose, and the pipeline funnel.

Reports: `CONSOLIDATED_REPORT.md` (the narrative synthesis) and this white-paper set (per-step deep dives with references).

4. Reproducibility and provenance

Every figure and table is a tracked artifact with code lineage, and every methodological choice is documented in the per-step white papers. Two caveats are carried consistently through every artifact: the screening library is ChEMBL-bioactivity-sourced with purchasability pending ZINC22 recovery; and STAT3's shallow SH2 pocket makes it a degrader/MD target rather than a classical-docking one.

5. How to run the packaged work

Each cluster package is self-contained (`environment.yml`, prepared inputs, SLURM scripts, README) and runs from a single `sbatch` after extraction: the docking package scales the screen across nodes; the co-folding package runs the Boltz-2 GPU cross-check on the top hits; and the MD package runs production from the pre-equilibrated systems and analyzes the trajectories.

6. Clinical framing

This is research and informational output, not clinical guidance. The leads and compounds identified are computational hypotheses that require experimental validation; therapeutic decisions must be made by qualified professionals with full patient context.

This white-paper set documents each pipeline step with references to the primary literature that established the methods and concepts used. All references are verified against CrossRef.

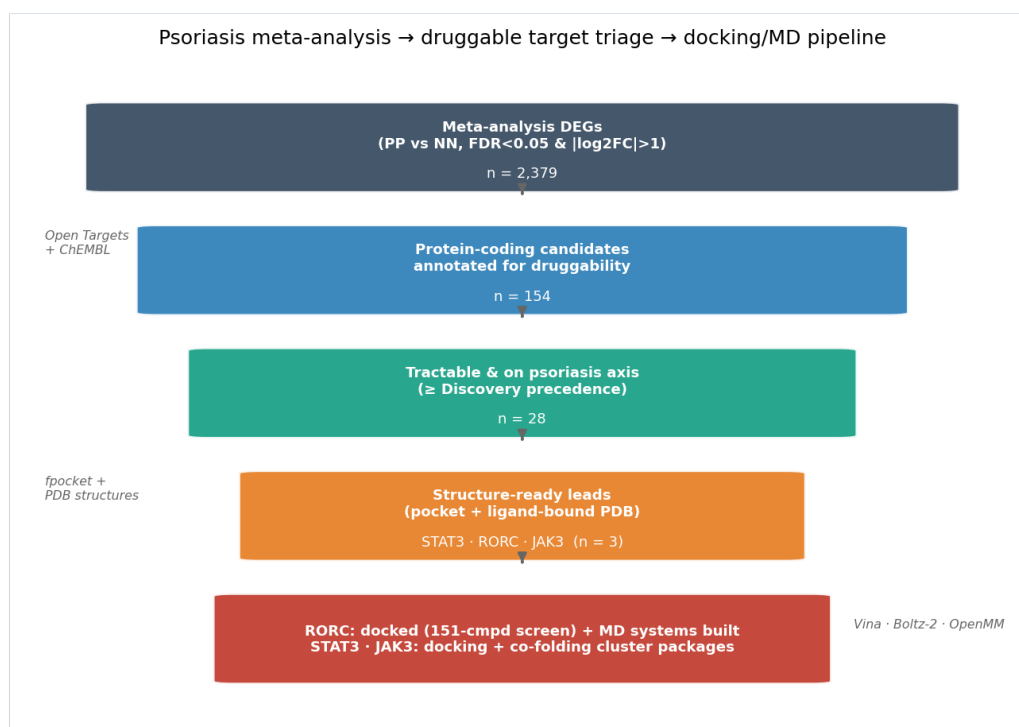


Figure: key output of this pipeline step.

References